SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)
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॥वसुधैव कुटुम्बकम्॥ Re-Accredited by NAAC with 'A++' Grade

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CSE A-2

Subject: Programming with Java
ASSIGNMENT No :3
TITLE : Developa Javaapplication demonstrating method overloading and the use of static variables,methods inclassdesign.

Problem Statement

Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Learning Objectives

To implement method overloading and use static variables and methods in Java programs.

Theory

Method overloading allows multiple methods to have the same name with different parameter lists, enabling compile-time polymorphism. The static keyword is used for variables and methods that belong to the class instead of individual objects. Static members share common data among all objects and reduce memory usage. These concepts improve code flexibility, readability, and efficient resource management.

```java
import java.util.*;

class Calculator {
    static int count = 0;

    static double add(double a, double b) {
        count++;
        return a + b;
    }

    static double add(double a, double b, double c) {
        count++;
        return a + b + c;
    }
}

public class Main1 {
    Run | Debug
    public static void main(String[] args) {
        Scanner in = new Scanner(System.in);

        System.out.print(s: "Enter first number : ");
        double a = in.nextDouble();

        System.out.print(s: "Enter second number : ");
        double b = in.nextDouble();

        System.out.print(s: "Enter third number : ");
        double c = in.nextDouble();

        System.out.println("\nSum of two : " + Calculator.add(a, b));
        System.out.println("Sum of three : " + Calculator.add(a, b, c));

        System.out.println("Number of calculations : " + Calculator.count);

        in.close();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp3> java Main1.java
Enter first number : 8
Enter second number : 4
Enter third number : 8

Sum of two : 12.0
Sum of three : 20.0
Number of calculations : 2
```

```java
class BankAccount {
    void rate () {
        System.out.println(x: "The ROI is 4%");
    }
}

class SavingsAccount extends BankAccount {
    void rate () {
        System.out.println(x: "The ROI is 6%");
    }
}

class CurrentAccount extends BankAccount {
    void rate () {
        System.out.println(x: "The ROI is 8%");
    }
}

public class Main2 {
    Run | Debug
    public static void main (String[] args) {
        CurrentAccount ca = new CurrentAccount();
        ca.rate();
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp3> java Main2.java
The ROI is 8%
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp3>
```

Exercise

1. Develop a **Calculator program** using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.

```java
import java.util.Scanner;

class Calculator {
    static int count = 0;

    int add (int a , int b) {
        count++;
        return a + b;
    }

    double add (double a, double b) {
        count++;
        return a + b;
    }
}

public class Ex1 {
    Run | Debug
    public static void main (String[] args) {
        try (Scanner in = new Scanner(System.in)) {
            System.out.print(s: "Enter first integer : ");
            int a = in.nextInt();

            System.out.print(s: "Enter second integer : ");
            int b = in.nextInt();

            System.out.print(s: "Enter first decimal number : ");
            double _a = in.nextDouble();

            System.out.print(s: "Enter second decimal number : ");
            double _b = in.nextDouble();

            Calculator addition = new Calculator();
            System.out.println("\nMethod One : Sum = " + addition.add(a, b));
            System.out.println("Method two : Sum = " + addition.add(_a, _b));
            System.out.println("Number of operations : " + Calculator.count);
        }
    }
}
```

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp3\Exercise> java Ex1.java
Enter first integer : 7
Enter second integer : 9
Enter first decimal number : 3.5
Enter second decimal number : 7.9

Method One : Sum = 16
Method two : Sum = 11.4
Number of operations : 2
```

2. Develop a **Restaurant Billing Application** where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

```
PS C:\Users\DHARMIT SHAH\Desktop\JAVA\Exp3\Exercise> java Ex2.java
------ Billing ------
Select Mode :
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 2
Enter the amount : Rs.8000
The total amount is = Rs.8800

------ Billing ------
Select Mode :
(1) Takeaway
(2) Dine-in
(3) Delivery
(4) Exit
Enter choice : 4

Exiting...

------ Total Orders ------
Total orders : 1
```

```java
import java.util.Scanner;

class RestaurantBilling {
    static int count = 0;

    // Takeaway
    int bill (double amount) {
        count++;
        return (int)(amount + 10);
    }

    // Dine-in
    int bill (double amount, double service) {
        count++;
        return (int)(amount + service / 100 * amount);
    }

    // Delivery
    int bill (double amount, double tax, double platform) {
        count++;
        return (int)(amount + platform / 100 * amount + tax / 100 * amount);
    }
}

public class Ex2 {
    Run | Debug
    public static void main (String[] args) {
        try (Scanner in = new Scanner(System.in)) {
            RestaurantBilling bill = new RestaurantBilling();

            while (true) {
                System.out.println(x: "------ Billing ------");

                System.out.println(x: "Select Mode : ");
                System.out.println(x: "(1) Takeaway");
                System.out.println(x: "(2) Dine-in");
                System.out.println(x: "(3) Delivery");
                System.out.println(x: "(4) Exit");
```

```java
            System.out.print(s: "Enter choice : ");
            int choice = in.nextInt();

            if (choice == 4) {
                System.out.println(x: "\nExiting...");
                break;
            }

            System.out.print(s: "Enter the amount : Rs.");
            double am = in.nextDouble();

            if (choice == 1)
                System.out.println("The total amount is = Rs." + bill.bill(am));
            else if (choice == 2)
                System.out.println("The total amount is = Rs." + bill.bill(am, service: 10
            else if (choice == 3)
                System.out.println("The total amount is = Rs." + bill.bill(am, tax: 2.5, p
            else
                System.out.println(x: "Wrong Choice!!!");

            System.out.println();
        }

        System.out.println();
        System.out.println(x: "------ Total Orders ------");
        System.out.println("Total orders : " + RestaurantBilling.count);
        System.out.println();
    }
  }
}
```